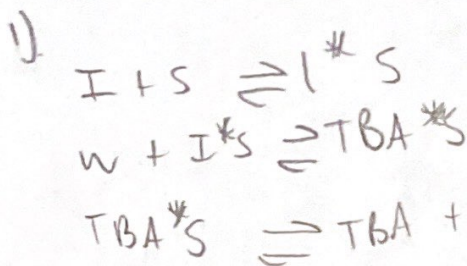


Cadmotel ~~Amendme~~ Exam 2.

1)

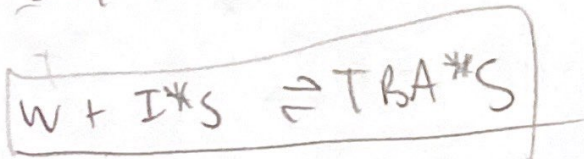


$$-r_{ad} = k_{ad} (C_I C_V - \frac{C_{IS}}{K_A})$$

$$\rightarrow -r_s = k_s (C_W C_{IS} - \frac{C_{TBA^*S}}{K_S})$$

$$\rightarrow -r_{de} = k_{de} (C_{TBA^*S} - \frac{C_{TBA} C_V}{K_D})$$

Eley Rideal mechanism: $A + S \rightarrow C + S$



$$K_S \ll K_A, K_D$$

$$\approx \text{no } \frac{r_d}{k_D} \approx \frac{r_A}{k_A}$$

$$\frac{-r_{ad}}{k_{ad}} = \frac{r_{de}}{k_D} \approx 0 \Rightarrow C_{TBA^*S}$$

$$\frac{-r_{ad}}{k_{ad}} = 0 = C_I C_V - \frac{C_{IS}}{K_A} \Rightarrow K_A C_I C_V = \frac{C_{IS}}{K_A}$$

$$\frac{r_{de}}{k_D} = 0 = C_{TBA^*S} - \frac{C_{TBA} C_V}{K_D} \Rightarrow C_{TBA^*S} = \frac{C_{TBA} C_V}{K_D}$$

$$r_s = k_s (C_{IS} C_W - \frac{C_{TBA^*S}}{K_S})$$

plugging in C_{IS} from above

$$r_s = k_s (K_A C_I C_V C_W - \frac{C_{TBA} C_V}{K_S \cdot K_D}) \Rightarrow C_V (K_A C_I C_W - \frac{C_{TBA}}{K_S \cdot K_D})$$

site balance

$$C_T = C_V + C_{IS} + C_{TBA^*S} = C_V (K_A C_I C_W + \frac{C_{TBA}}{K_D} + 1)$$

$$C_V = \frac{C_T}{(1 + K_A C_I C_W + \frac{C_{TBA}}{K_D})}$$

+

$$r_s = \frac{k_s (K_A C_I C_W - \frac{C_{TBA}}{K_D \cdot K_S}) C_T}{(1 + K_A C_I C_W + \frac{C_{TBA}}{K_D})}$$

2). $A \rightarrow B$, first order.

$$k = 6.6 \times 10^{-3} \text{ min}^{-1}$$

$$T_0 = 350 \text{ K}$$

$$E = 40000 \frac{\text{cal}}{\text{mol} \cdot \text{K}}$$

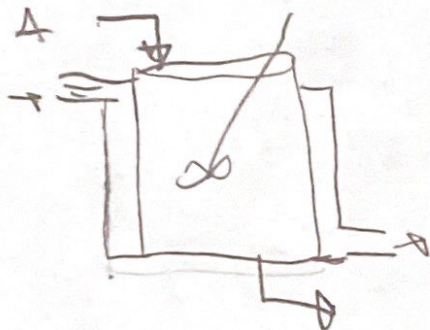
exothermic

CSTR

Heat Specs

$$UA = 8000 \frac{\text{cal}}{\text{min} \cdot ^\circ\text{C}}$$

$$T_a = 300 \text{ K}$$



A initial conditions

$$F_{A0} = 80 \frac{\text{mol}}{\text{min}}$$

$$\tau = 100 \text{ min}$$

$$C_{pA} = 20 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}}$$

I initial conditions

$$F_{I0} = F_{A0} = 80 \frac{\text{mol}}{\text{min}} \quad \text{equimolar}$$

$$C_{pI} = 30 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}}$$

$$\Delta H_{rxn} = -7500 \frac{\text{cal}}{\text{mol}} = \text{exothermic}$$

negative

a) feed Temp 450 K - reactor temp?

$$G(T) = - \frac{\Delta H_{rxn} + T K}{1 + T K}$$

heat at into system.

$$k = k(350 \text{ K}) \times e^{\left[\frac{E}{R} \left(\frac{1}{350 \text{ K}} - \frac{1}{T} \right) \right]} = k = 6.6 \times 10^{-3} \text{ min}^{-1} e^{\left[\frac{40000 \frac{\text{cal}}{\text{mol} \cdot \text{K}}}{2 \text{ cal/mol} \cdot \text{K}} \left(\frac{1}{350 \text{ K}} - \frac{1}{T} \right) \right]}$$

$$Q(T) = \bar{C}_{p0} (1 + K) (T - T_c), \quad K = \frac{UA}{C_{p0} F_{A0}} = \frac{8000 \frac{\text{cal}}{\text{min} \cdot ^\circ\text{C}}}{(1 \times 30 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}} + 1 \times 20 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}}) 80 \frac{\text{mol}}{\text{min}}}$$

$$K = \frac{8000 \frac{\text{cal}}{\text{min} \cdot ^\circ\text{C}}}{4000 \frac{\text{cal}}{\text{min} \cdot ^\circ\text{C}}} = 2 = K$$

$$T_c = \frac{k T_a + t_0}{1 + K} = \frac{2(300 \text{ K}) + 450 \text{ K}}{1 + 2} = \frac{1050 \text{ K}}{3} = 350 \text{ K} = T_c$$

$$\bar{C}_{p0} (1 + K) (T - T_c) = \left(1 + 30 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}} + 1 \times 20 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}} \right) 3 (T - 350 \text{ K}) = 50 \text{ cal} \cdot ^\circ\text{C}$$

$$= 150 \frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}} (T - 350 \text{ K})$$

$$G(T) = R(T) \Rightarrow \frac{\Delta H_r T_c}{1 + k_c} = C_{A0}(1+k)(T-T_c) = \cancel{K} =$$

$$\frac{-7500 \frac{\text{cal}}{\text{mol}} (100 \text{ min}) \left[6.6 \times 10^3 \text{ min}^{-1} e \left[20000 k \left(\frac{1}{330K} - \frac{1}{T} \right) \right] \right]}{1 + 100 \text{ min} (K^{\phi})} = 150 \frac{\text{cal}}{\text{mol}^{\circ}\text{C}} (T - 330K)$$

$$4950 \frac{\text{cal}}{\text{mol}} e \left[20000 k \left(\frac{1}{330K} - \frac{1}{T} \right) \right]$$

$$= 150T - 52500$$

$$\frac{1 + 0.66 e \left[20000 k \left(\frac{1}{330K} - \frac{1}{T} \right) \right]}{152500 - 150T}$$

$$= 0 \Rightarrow$$

$$152500 - 150T$$

I'm using excel... pls don't think I'm cheating...

I'm using goal seek for $T = 400$ because it's exothermic.
 T should be between Jacket & Feed. Maybe bigger.

$$T = 391.9397^{\circ}\text{K} \approx 400^{\circ}\text{K}$$

b) plot reactor T as $f(T_{\text{Feed}})$ from $310K$ to $430K$.

$G(T)$ = same as above. - k constant

$$R(T) = (150)(T - T_c) \xrightarrow{T_c = \frac{k T_{\text{act}} T_0}{k + 1}} = 150 \frac{\text{cal}}{\text{mol}^{\circ}\text{C}} (T - T_c)$$

$$G(T) = L(T).$$

$$4955 \frac{\text{cal}}{\text{mol}} e^{20000 K \left(\frac{1}{350} - \frac{1}{T} \right)}$$

$$1 + 66 e^{20000 K \left(\frac{1}{350} - \frac{1}{T} \right)}$$

constant.

$$G(T) - 150T + 150T_c = 0$$

$$G(T) - 150T + 150 \left[\frac{600 + T_0}{3} \right] = 0.$$

$$-150(T - T_c)$$

$$= 150T - 150T_c = 150(T - T_c)$$

At this point, I assume
you expect us to use
excel.... Pls!!!

First guess for $T_0 = 310 \text{ K}$, $T = 305 \text{ K}$ between feed & heat exchanger. (see appendix).

I'll use for $T_0 = 310 \text{ K}$, $T = 303.3 \text{ K}$.

I'll include b) plot in back (see appendix B).

$$c). G(T) = - \frac{\Delta H_{rx} X}{T_{ref} K} = \frac{\Delta H_{rx} K}{T_{ref} K} X_{mb} = \frac{T_c}{T_{ref} K}.$$

Taking T 's from Part B (T_0 & T) which are at
Energy equilibrium, $K(T)$

solving for x . (See Appendix

High conversion begins between $T = 340$ & 400 K .

$$\begin{aligned} T &= 390 \text{ K} \\ x &= .463 \\ T &= 400 \text{ K} \\ x &= .408 \end{aligned}$$

d) at $T_0 = 400^\circ$
 Reader $T = 382.771$ $+5^\circ$
 $387.772 - 20^\circ = 367.772 = T_0 - \text{fluid.}$

T_0
 350
 370
 interpolating
 $.12351$

fluid inlet	X
T_0	
360	.0032
370	.0064

fluid temp.
 $T_0 = 367.7712$
 $x = .00572$

e). inlet extinction T 's $\sim 380 \sim 390$ K for this backm

3) Initially, I did a mol balance & eqn equation in terms of x . when I combined them, I got $\frac{dx}{dw} = k \frac{v_0}{C_{A0}} \frac{(1-x)^2}{(1+Kx)^2} [1 - (dx)(x)w]$

(The work will be in the appendix). I figured I could separate variables,

integrate and solve for w where $x=0.99$. Then I could find length of the PBR because $w = \rho_c A_c z (1-\phi)$ where z will be the length of the reactor. But then I stopped because

I realized I was missing parameters & example Dr Berry posted used a ϕ catalysis-equations specific approach such as solving for η & ϕ . I read that micro-PBRs are very affected by pressure drop, in the beginning of the chapter for pressure drop. However I don't see the pressure drop being accounted for in Dr Berry's approach which brings us here --

In ch 14 we saw that for high velocity $\Omega \approx \eta$. & we learned that a high pressure drop leads to an increase in flow rate, compared to no pressure drop. And we know there will be pressure drop in this micro-PBR.

so we can write $-r_A' = -\Omega k' C_A^2$ & back to PBR mol balance

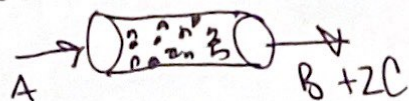
$$-F_{A0} \frac{dx}{dw} = \Omega k' C_{A0} (1-x)^2$$

$$\ln(1-x)^2 = -\frac{\Omega k' C_{A0}^2 w}{F_{A0}}$$

& solve for w , then z ...?

-because Ω is overall effectiveness it will account for parameters such as gas phase having more catalyst contact & reacting more. I hope...? so I don't use $(1+Kx)$...?

3) - micro-PBR



Appendix Part of 3



$$2^{\text{nd}} \text{ order} \Rightarrow -r_A = k C_A^2$$

Assumptions

- Steady state.
- Isothermal because no info on thermal or chemical identities. Also "micro" probably not big enough for ΔT .
- Gas

Mol balance

$$F_{A0} \frac{dx}{dW} = -r_A \Rightarrow \frac{dx}{dW} = \frac{-r_A}{F_{A0}}$$

isothermal.

$$\text{ergun} \quad \frac{dp}{dW} = \frac{-\alpha}{2p} \left(\frac{F_{A0}}{F_{A0}} \right) \Rightarrow \frac{dp}{dW} = \frac{-\alpha}{2p} (1 - \epsilon x)$$

convert

Converting to Conversion

$$p = \frac{P}{P_0}$$

$$F_{A0} = C_{A0} V_0$$

$$C_A = \frac{F_A}{V} = C_{A0} \frac{(1-x) \left[\frac{P}{P_0} \right]}{(1+\epsilon x) \left[\frac{P}{P_0} \right]} \Rightarrow p$$

$$\frac{dx}{dW} = \frac{-r_A}{F_{A0}} = \frac{dx}{dW} = k' \frac{C_{A0}^2 (1-x)^2}{F_{A0} (1+\epsilon x)} \left(\frac{P}{P_0} \right)^2$$

in at $\frac{1}{h_1}$

ergun

$$\frac{dp}{dW} = \frac{-\alpha}{2p} (1 - \epsilon x) \Rightarrow \int p dp = \int \frac{-\alpha}{2} (1 - \epsilon x) dW$$

$$p = \frac{P}{P_0}$$

$$\int p dp = \frac{-\alpha}{2} (1 - \epsilon x) W \quad \leftarrow W=0 \text{ at inlet}$$

$$\Rightarrow \frac{p^2}{2} - 1 = \frac{-\alpha}{2} (1 - \epsilon x) W$$

$$\left(\frac{p}{2} \right) = 1 - \frac{\alpha}{2} (1 - \epsilon x) W$$

plug back into mol balance.

$$\frac{dx}{dw} = k' \frac{C_{A0}^2 (1-x)^2}{F_{A0} (1+\theta)^2} \cdot [1 - 2(1-x)W]$$

$$W = (1-\phi) A_c \cdot Z$$

$$\frac{dx}{dw} = k' \frac{C_{A0}^2}{F_{A0}} \frac{(1-x)^2}{(1+\theta)^2} [1 - 2(1-x)W]$$

APPENDIX

2a-

	A	B	C
1	2(a)		
2	T =	400	
3	const Term	1265.038	
4	Equation	-8.9721	
5			
6	Goal Seek ? X		
7	Set cell:	\$B\$4	↑
8	To value:	0	
9	By changing cell:	\$B\$2	↑
10	OK	Cancel	

	A	B
1	2(a)	
2	T =	399.9397
3	const Term	1255.542
4	Equation	-4.9E-05
5		
6		
7		
8		

2b

	A	B	C
1	2(a)		
2	T =	305	
3	const Term	0.000218	
4	G(T) - [150(T-Tc)]	-248.921	
5	Tc	303.3333	
6	T0	310	
7			
8	Goal Seek ? X		
9	Set cell:	\$B\$4	↑
10	To value:	0	
11	By changing cell:	\$B\$2	↑
12	OK	Cancel	

	A	B
1	2(a)	
2	T =	303.3384
3	const Term	0.000152
4	G(T) - [150(T-Tc)]	-6E-06
5	Tc	303.3333
6	T0	310
7		
8		

T0	T	X
310	303.3333	0.0001
320	306.677	0.000206
330	310.0208	0.000416
340	313.3748	0.000829
350	316.7484	0.001635
360	320.16	0.0032
370	323.6459	0.006252
380	327.2842	0.012351
390	350.1257	0.402514
400	382.7712	0.988758
410	386.3172	0.993011
420	389.7788	0.995576
430	393.1914	0.997161

